



*It Appears You have Eaten a Bee*

*Please write your name in the box below.*

## **Basic Algebra I Midterm 1**

**31 January 2025, 12.30 PM-1.20 PM**

| <i>Weight</i> | <i>Problem</i> |
|---------------|----------------|
| 5             | 1(i)           |
| 5             | 1(ii)          |
| 10            | 1(iii)         |
| 5             | 2(i)           |
| 5             | 2(ii)          |
| 10            | 2(iii)         |
| 10            | 3              |
| 50            |                |

**Problem 1.** Let  $X$  and  $Y$  be two sets, and let  $f : X \rightarrow Y$  be a function.

(i) What does it mean to say that a function  $Y \rightarrow X$  is *inverse* to  $f$ ?

(ii) What does it mean to say that  $f$  is *bijective*?

- (iii) Let  $Z$  be another set, and let  $g : Y \rightarrow Z$  be another function. Suppose that  $f$  and  $g$  are bijective. Show directly from the definitions that  $g \circ f$  is bijective.  
(Do *not* use that bijective  $\Leftrightarrow$  injective and surjective.)

**Problem 2.**

- (i) Let  $\sim$  be an equivalence relation on a set  $X$ , and write  $\pi : X \rightarrow X/\sim$  for the quotient map. Let  $Y$  be another set, and let  $F : X \rightarrow Y$  be a function. State the criterion on  $F$  for there to exist a function  $f : X/\sim \rightarrow Y$  satisfying

$$f \circ \pi = F.$$

- (ii) Let  $n$  be an integer. We defined the set of integers mod  $n$  to be  $\mathbf{Z}/n\mathbf{Z} = \mathbf{Z}/\sim_n$  for a certain equivalence relation  $\sim_n$  on  $\mathbf{Z}$ . Give the definition of  $\sim_n$ .  
(You do not need to show that  $\sim_n$  is an equivalence relation.)

(iii) Let  $m$  and  $n$  be integers. Use the criterion from (i) to construct a function

$$\mu : \mathbf{Z}/n\mathbf{Z} \rightarrow \mathbf{Z}/mn\mathbf{Z}$$

satisfying<sup>1</sup>

$$\mu([a]_n) = [m \cdot a]_{mn} \quad \text{for all } a \in \mathbf{Z}.$$

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<sup>1</sup>Here  $[a]_n$  (resp.  $[a]_{mn}$ ) denotes the image of  $a$  under the quotient map  $\mathbf{Z} \rightarrow \mathbf{Z}/n\mathbf{Z}$  (resp.  $\mathbf{Z} \rightarrow \mathbf{Z}/mn\mathbf{Z}$ ).

**Problem 3.** Find an element  $x \in \mathbf{Z}/23\mathbf{Z}$  satisfying

$$[16] \cdot x = [1].$$